UI GreenMetric, Wageningen University & Research, and UBB

Team: Moldovan Mihai, Grec Marc-Alexander

The UI GreenMetric ranking

UI GreenMetric, launched in 2010 by Universitas Indonesia, was the first world university ranking dedicated to campus sustainability. The 2025 edition evaluated **1,745 universities from 105 countries** [1]. Institutions self-report on six weighted criteria:

Setting & Infrastructure	Energy & Climate Change	Waste	Water	Transportation	Education & Research
15%	**21%**	18%	10%	18%	18%

Energy & Climate Change is the heaviest criterion; its indicators include **smart building implementation**, energy-efficient appliances, renewable energy usage and production, **electricity use per campus inhabitant** and carbon footprint per inhabitant [2].

Selected university: Wageningen University & Research (WUR), the Netherlands

WUR ranked **1st worldwide in 2025**, and was also best in Europe and best in the Waste and Water criteria [1].

Challenges. WUR is a research-intensive university with energy-hungry laboratories, greenhouses and animal facilities, where research must never be interrupted by a power cut; renewable supply is intermittent; and energy must remain affordable [3].

Solutions. WUR owns **26 wind turbines** in Lelystad; together with solar panels, its own sustainable generation exceeded its total energy consumption in 2020 (121%) [4]. It has put about **10,000 solar panels** on its roofs [4], is connecting campus buildings to **aquifer thermal energy storage (ATES)** with heat pumps to phase out natural gas [3, 4], and saves energy through insulation, LED lighting and **optimised building management systems** that automatically adapt installations to conditions [3].

UBB: where it stands

UBB has been the **best-ranked Romanian university** in GreenMetric for years. It climbed steadily from 340th (2018) to **122nd (2024)**, then fell to **287th of 1,745 in 2025**, while remaining 1st nationally [5, 6]. Its reports show clear strengths: more than 35% of campus area is covered by forest vegetation, selective waste collection is an institution-wide policy, and budget students receive 120 free public-transport trips per month [7].

Challenges identified in the UBB reports [5, 7]:

- **Energy is the weak spot.** The 2023 report gives no energy, water or waste baselines; solar panels were still only at the "identify opportunities" stage; only **2 smart buildings** and 1 green building are reported.
- Several measures depend on available funding, and some deadlines (composting, green roofs) slipped.
- The 2025 drop shows that the lead is not secure as more universities enter the ranking.

Strategies for improvement at UBB

UBB's draft Climate Action Plan already sets ambitious targets: from 2 to **at least 5 smart buildings by 2028**, **motion sensors in at least 50% of common spaces**, per-capita electricity **below 250 kWh** (from under 279 kWh), at least 500 kW of solar by 2028 and a 5% cut in total energy use [5]. Learning from WUR, we propose:

1. **Occupancy-aware smart rooms.** Retrofit classrooms with low-cost controllers that switch lighting off and heating to an economy setpoint when rooms are empty, coordinated by a central building energy manager. This directly scores on the *smart building* and *electricity per capita* indicators.
2. **Measure before optimising.** Per-building and per-room metering with a public dashboard, so that every target in the action plan has a baseline and a measurable trend.
3. **Rooftop solar plus demand shifting**, scheduling flexible loads when on-site generation is high.
4. **Close the open items** (composting, water-saving fixtures) that keep Waste and Water points low.

Proposal 1 is the starting point for our lab project (see the problem statement).

Sources. [1] UI GreenMetric, *2025 results and award ceremony*, greenmetric.ui.ac.id [2] UI GreenMetric methodology (criteria and indicators) [3] WUR, *Energy*, wur.nl [4] WUR, *From carbon energy… to carbon neutral*, wur.nl [5] UBB, *Climate Action Plan* (draft revision), green.ubbcluj.ro [6] Agerpres, 22 Dec 2025 [7] UBB, *Raport de dezvoltare durabilă 2023*, green.ubbcluj.ro